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COM3502/4502/6502
SPEECH PROCESSING

Lecture 20

/ faɪnəl əʊvəvju: /

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This module has covered ...

- Speaking
- Hearing
- The nature of speech
- Sounds and symbols
- Phonetics
- Signals and spectra
- Sampling
- Waveform processing
- The Fourier and Z transforms
- Filters and linear prediction
- Cepstral analysis

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So, what have we learnt?

... about speech

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The Speech Chain

The diagram illustrates the speech chain from a speaker to a listener. On the left, the **SPEAKER** section shows a **BRAIN** connected to **Sensory Nerves** and **Motor Nerves**. The **Motor Nerves** lead to the **Vocal Muscles**, which produce **Sound Waves**. A **Feedback Link** is shown as a curved arrow from the speaker's ear back to their brain. On the right, the **LISTENER** section shows **Sound Waves** entering the **Ear**, which connects to **Sensory Nerves** and then to the **BRAIN**. A large red arrow highlights the path of sound waves from the speaker's vocal muscles to the listener's ear. Below the diagram, a horizontal flowchart shows the levels of processing: **LINGUISTIC LEVEL** → **PHYSIOLOGICAL LEVEL** → **ACOUSTIC LEVEL** → **PHYSIOLOGICAL LEVEL** → **LINGUISTIC LEVEL**.

Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

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Speech Segments

The collage features several linguistic and phonetic resources. On the left is the International Phonetic Alphabet (IPA) chart, showing various consonant and vowel symbols. To its right is a sagittal cross-section of the human head, highlighting the vocal tract. Further right is a diagram titled 'ELEMENTS OF ACUSTIC PHONETICS' showing waveforms for different speech sounds. In the center is a photograph of a seal, with the caption 'Hoover the seal' below it.

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The cartoon depicts two women at a sandwich stand. The woman on the left, with blonde hair, is thinking, "It's lunchtime. I wonder if she'd like one of my sandwiches?". The woman on the right, with dark hair, is speaking, and her speech is represented by a complex phonetic transcription: [sɪlɒntʃtʌɪm | dʒəlaɪk wʌnəməsæmwədʒɪz]. A clock icon is visible in the background.

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Words-Syllables-Phonemes

“It is lunchtime. Would you like one of my sandwiches?”

Orthographic Transcription

VC VC CVCC-CVC CVC CV CVC CVC VC CV CVCC-CVC-VC

CV Skeleton

/ ɪt ɪz lʌntʃtaɪm || wʊd ju: laɪk wʌn ʌv maɪ sændwɪtʃɪz /

Phonemic Transcription (IPA)

/ ɪt ɪz lʌntʃtaɪm || wʊd ju: laɪk wʌn ʌv maɪ sʌndwɪtʃɪz /

Phonemic Transcription (SAMPA)

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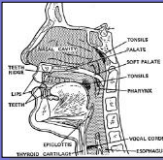


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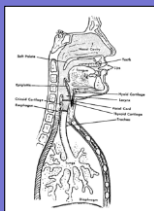
VPM Labels & Vowel Qualities



“sandwiches” → [sæmwədʒɪz]

- [s] voiceless alveolar fricative
- [æ] open front vowel
- [m] voiced bilabial nasal
- [w] voiced labial-velar approximant
- [ə] neutral vowel (*schwa*)
- [dʒ] voiced postalveolar affricate
- [ɪ] half-close front vowel
- [z] voiced alveolar fricative

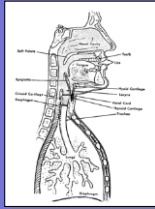
Articulatory Sequence



“sandwiches” → [sæmwədʒɪz]

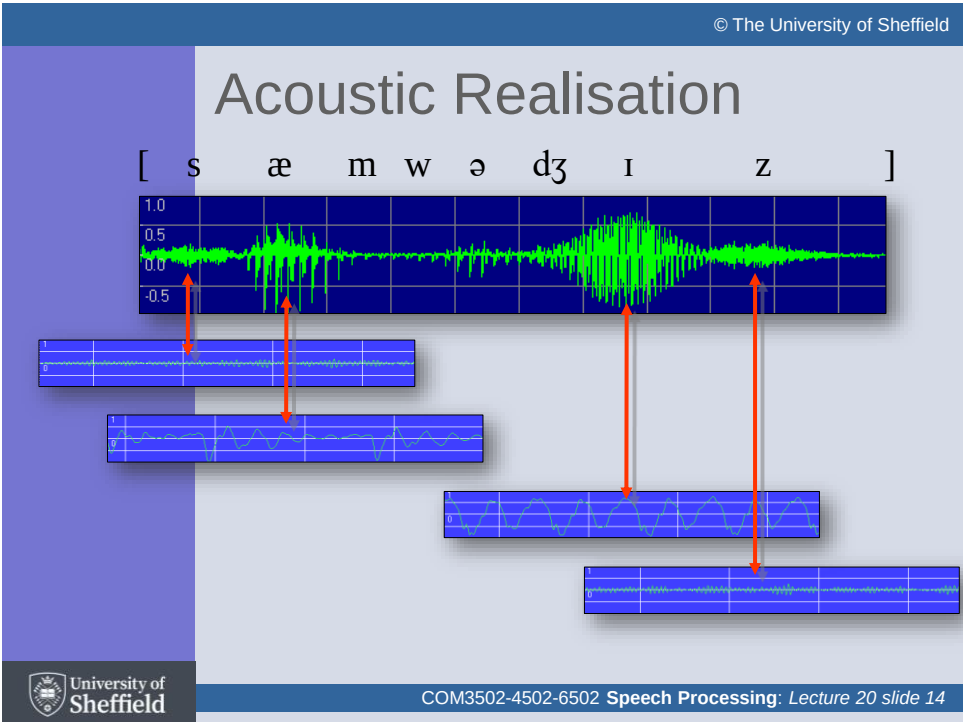
- ingress of air into the lungs (*breath in*)
- raise blade of tongue to the alveolar ridge forming a narrow channel
- pressure from the lungs initiates the air flow
- air flow through the narrow opening between the tongue blade and the alveolar ridge generates high frequency friction: [s]
- tongue moves away from the roof of the mouth to an open front position
- the vocal cords start to vibrate: [æ]
- lips come together to create a bilabial closure
- velum is open
- air flow continues through the nasal cavity: [m]

Articulatory Sequence (cont.)

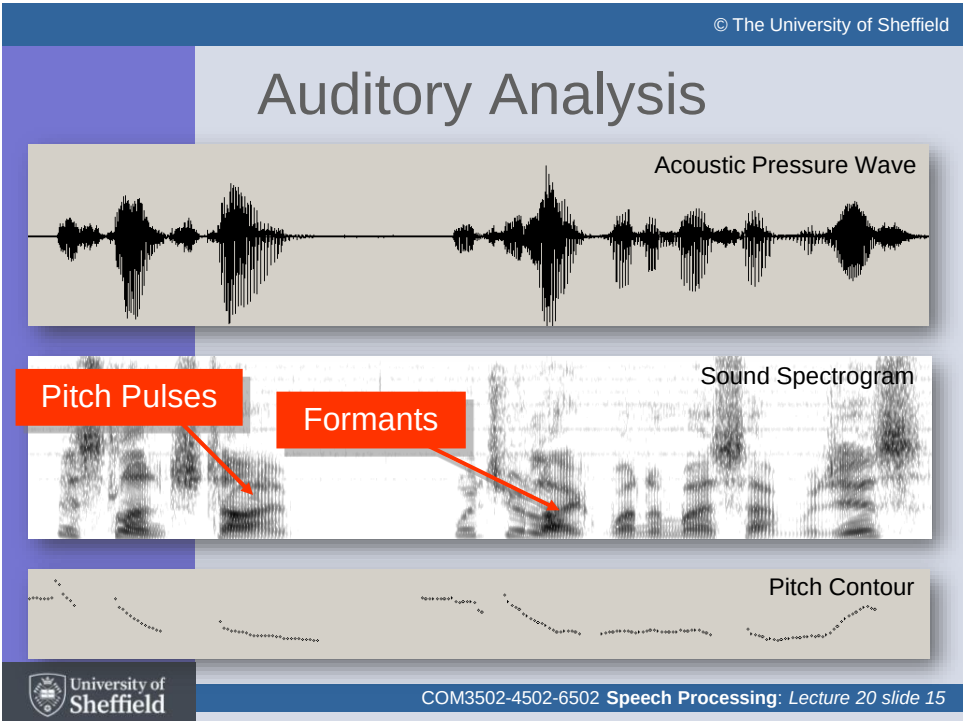


“sandwiches” → [sæmwədʒɪz]

- lips move to a rounded position and open: [w]
- tongue moves to a neutral position: [ə]
- velum closes
- tongue blade moves to the postalveolar position and forms a closure: [dʒ]
- pressure builds in the oral cavity
- tongue moves slowly away from its postalveolar position releasing the closure and generating a release of air
- tongue moves to a half-close front position: [ɪ]
- tongue blade moves to the alveolar ridge and forms a narrow channel
- air flow through the narrow opening between the tongue blade and the alveolar ridge generates high frequency friction: [z]
- vocal cords stop vibrating
- air flow stops



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Segmentation

A spectrogram of the sentence "I am a sandwich" is shown. The spectrogram is divided into segments by vertical blue lines. Two red arrows point to specific segments: one to the 's' segment labeled "Phonetic Segmentation" and another to the 'dʒ' segment labeled "Acoustic Segmentation". Below the spectrogram, the phonetic transcription [s æ m w ə dʒ ɪ z] is displayed.

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A cartoon illustration shows a woman in a red shirt handing a sandwich to a man in a purple shirt. The woman's speech bubble contains the phonetic transcription [sɪlɒntʃaɪm I dʒələɪk wʌnəməsæmwədʒɪz]. The man's thought bubble contains the text "She's asking me if I'd like one of her sandwiches". A clock icon is shown in the top right corner.

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['æʃlɪ ɪ
'fædmələntʃə:ɪdi'θæŋks
ɪ ænəv ɡɒtə ɡəʊn 'mi:t
sɑmwaɪn ɪ 'kænəbɒrəʒə
'fəʊn]

Oh dear, I need to go – but I've forgotten my phone

Actually, I had my lunch already, thanks. And I got to go and meet someone. Can I borrow your phone?



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['sʊə 'hiəʒəɡəʊ ɪ
ɪt ju:zəz 'dʒi:'es'em ju:nəʊ]

GSM?

Sure, here you go. It uses GSM, you know.



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What have we learnt?

... about speech processing

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Digital Speech Coding

ETSI
GSM
TECHNICAL
SPECIFICATION

GSM 03.00
August 2000
Version 5.1.0

Serial: 470 000

Key words: Digital cellular telecommunication system; Channel Coding for digital communication; GSM

ICS: 33.020

Key words: Digital cellular telecommunication system; Channel Coding for digital communication; GSM

GSM
GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS

Digital cellular telecommunication system (Phase 2+):
Handover procedures
(GSM 03.00 version 5.1.0)

ETSI

European Telecommunications Standards Institute

470 Boulevard
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European Telecommunications Standards Institute

Google Meet

zoom

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Digital Speech Coding



- 5 min studio-quality audio recording
 - 24 bits/sample
 - 48 kHz sampling rate
- Dynamic range
 - = $20 \log_{10}(2^{24}-1)$
 - = **144 dB**
- File size
 - = $48,000 \times 24 \times 5 \times 60 / 8$
 - = 43,200,000
 - = **43.2 Mbytes**
- Streaming rate
 - = $48,000 \times 24$
 - = 1,152,000
 - = **1.152 Mbps**

Equivalent to 88 simplex GSM channels

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Speech Analysis



$\frac{4 \sin \theta}{\pi}$

$\frac{4 \sin 3\theta}{3\pi}$

$\frac{4 \sin 5\theta}{5\pi}$

$\frac{4 \sin 7\theta}{7\pi}$

$\frac{2 \sin \theta}{-\pi}$

$\frac{2 \sin 2\theta}{2\pi}$

$\frac{2 \sin 3\theta}{-3\pi}$

$\frac{2 \sin 4\theta}{4\pi}$

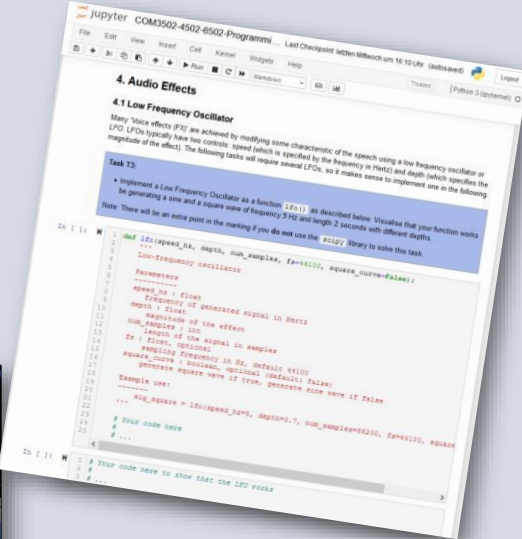
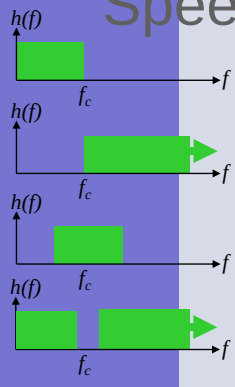
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Speech can be Manipulated



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What have we learnt?

... about the future

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Speech Processing in the Future?

<https://youtu.be/3I24bSteJpw>

Introducing Translate for Animals
github.com/translateforanimals

<https://youtu.be/jABwjIu2jFY>

RICH PEOPLE PROBLEMS

https://youtu.be/MaNU_Ps-eRU

<https://youtu.be/6MzmIUwFzds>

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So, that's it!

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- The nature of speech
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- Phonology
- Prosody
- Signals and spectra
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The Assignment and Exam

- Assignment
 - Due next Friday!
 - Dec. 15th 2023
 - Submission via Blackboard
- Exam:
 - Info from <https://students.sheffield.ac.uk/exams/timetables/#autumn-semester-timetables>
 - 29 January 2024
 - Diamond, Computer Room 1
 - 9.00 - 11.30 pm (2.5 hours time)
 - Open book
 - Access to Blackboard ✓
 - simple calculator allowed, see <https://students.sheffield.ac.uk/exams/rules/calculator-checker> ✓
 - No direct Internet, no phones, no own devices, no contact with others, no paper notes, etc. ✗

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Any Questions ?



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(they take less than 10 minutes)



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Thanks & Goodbye

/ ænd gud lʌk wið ðə əsaɪnmənt ənd
ði egzæm /



https://youtu.be/g_imiTxXHQ



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